# Single-User, Single-Target Robust Hybrid ISAC with Fluid Transmit Antennas

**Scope.** This proposed model retains the robust hybrid-beamforming formulation of [1], specializes it to one user and one target, and replaces only the fixed transmit ULA coordinates by adjustable one-dimensional fluid-antenna coordinates. The radar receive array remains a fixed ULA. Continuous spatial reconfiguration is motivated by [2]; we do not adopt that paper's different sensing objective or fluid user antenna.

## 1 System Model

### 1.1 Network and antenna geometry

Consider a downlink integrated sensing and communication (ISAC) system with a dual-function base station (BS), one fixed single-antenna communication user, one point-like target, and $J \geq 0$ clutter sources. The BS employs $N_t$ fluid transmit antennas, $L$ RF chains, and $N_r$ fixed radar receive antennas. We consider narrowband far-field operation. Antenna locations are configured before a transmission block and held fixed during that block. The array aperture is small relative to propagation distances, so path angles and gains are held fixed as candidate transmit positions are evaluated.

Let $\mathbf{p} = [p_1, \ldots, p_{N_t}]^T \in \mathbb{R}^{N_t}$ collect the transmit coordinates on an aperture of length $A$. Label the antennas in spatial order and define

$$\mathcal{P} = \{\mathbf{p} : \ 0 \leq p_1, \quad p_{n+1} - p_n \geq d_{\min}, \ n = 1, \ldots, N_t - 1, \quad p_{N_t} \leq A\}. \tag{1}$$

Here $d_{\min} > 0$ is the minimum antenna separation. The necessary and sufficient condition for this geometric set to be nonempty is $A \geq (N_t - 1)d_{\min}$. This ordered one-dimensional set is convex; the complete optimization problem will nevertheless be nonconvex. We assume accurate position control and neglect mutual coupling in this initial model. The receive ULA has spacing $d_r$.

### 1.2 Position-dependent transmit response

Let $\lambda$ be the wavelength, $k_0 = 2\pi/\lambda$, and $\theta$ denote an angle measured from array broadside. Following the phase convention in [1], the nominal transmit steering vector is

$$\widehat{\mathbf{a}}_t(\theta; \mathbf{p}) = \left[e^{-jk_0 p_1 \sin\theta}, \ldots, e^{-jk_0 p_{N_t} \sin\theta}\right]^T. \tag{2}$$

Indeed, the relative path-length term at antenna $n$ is $p_n \sin\theta$; multiplication by $k_0$ gives its relative phase. For the fixed receive ULA,

$$\widehat{\mathbf{a}}_r(\theta) = \left[1, e^{-jk_0 d_r \sin\theta}, \ldots, e^{-jk_0 (N_r - 1) d_r \sin\theta}\right]^T. \tag{3}$$

Setting $p_n = (n-1)d$ in (2) recovers the original transmit ULA with spacing $d$. Thus the fixed-array geometry is a special case whenever it satisfies (1).

### 1.3 Position-dependent communication channel

To evaluate different geometries consistently, model the nominal communication channel through $K_c$ propagation paths:

$$\widehat{\mathbf{h}}^H(\mathbf{p}) = \sum_{k=1}^{K_c} \beta_k \widehat{\mathbf{a}}_t^T(\vartheta_k; \mathbf{p}), \qquad \widehat{\mathbf{h}}(\mathbf{p}) \in \mathbb{C}^{N_t}. \tag{4}$$

Here $\beta_k$ includes the complex path gain at the aperture reference and $\vartheta_k$ is its departure angle. Consequently, $[\widehat{\mathbf{h}}^H(\mathbf{p})]_n = \sum_k \beta_k e^{-jk_0 p_n \sin\vartheta_k}$. The gains and angles are fixed within each scenario; channels must not be independently redrawn after position updates. The actual channel is

$$\mathbf{h}(\mathbf{p}) = \widehat{\mathbf{h}}(\mathbf{p}) + \boldsymbol{\delta}_h. \tag{5}$$

This spatial construction is the channel-model addition required to make fluid positioning meaningful; the received-signal structure remains unchanged.

### 1.4 Hybrid transmit signal

Let $z_c$ and $z_s$ denote the zero-mean communication symbol and dedicated sensing symbol. Assume

$$\mathbb{E}|z_c|^2 = \mathbb{E}|z_s|^2 = 1, \qquad \mathbb{E}[z_c z_s^*] = 0. \tag{6}$$

The analog beamformer is $\mathbf{F}_a \in \mathbb{C}^{N_t \times L}$ and the digital beamformer is $\mathbf{F}_d = [\mathbf{f}_c, \mathbf{f}_s] \in \mathbb{C}^{L\times 2}$. For the initial two-stream implementation, choose $2 \le L \le N_t$. The transmitted signal is

$$\mathbf{x} = \mathbf{F}_a\mathbf{f}_c z_c + \mathbf{F}_a\mathbf{f}_s z_s = \mathbf{F}_a\mathbf{F}_d\mathbf{z}, \qquad \mathbf{z} = [z_c, z_s]^T. \tag{7}$$

Writing $\mathbf{v}_c = \mathbf{F}_a\mathbf{f}_c$ and $\mathbf{v}_s = \mathbf{F}_a\mathbf{f}_s$, the transmit covariance and average radiated power follow as

$$\mathbf{R}_x = \mathbb{E}[\mathbf{x}\mathbf{x}^H] = \mathbf{v}_c\mathbf{v}_c^H + \mathbf{v}_s\mathbf{v}_s^H = \mathbf{F}_a\mathbf{F}_d\mathbf{F}_d^H\mathbf{F}_a^H, \tag{8}$$

$$\mathbb{E}\|\mathbf{x}\|_2^2 = \mathrm{Tr}(\mathbf{R}_x) = \|\mathbf{F}_a\mathbf{F}_d\|_F^2. \tag{9}$$

The cross terms vanish by the uncorrelated-stream assumption. We retain the original constraints $|[\mathbf{F}_a]_{n,\ell}| = 1$ and $\|\mathbf{F}_a\mathbf{F}_d\|_F^2 = P_{\mathrm{BS}}$.

### 1.5 Communication received signal and rate

With noise $n_c \sim \mathcal{CN}(0, \sigma_c^2)$, the user receives

$$y_c = \mathbf{h}^H(\mathbf{p})\mathbf{x} + n_c = \underbrace{\mathbf{h}^H(\mathbf{p})\mathbf{F}_a\mathbf{f}_c z_c}_{\text{desired signal}} + \underbrace{\mathbf{h}^H(\mathbf{p})\mathbf{F}_a\mathbf{f}_s z_s}_{\text{sensing interference}} + n_c. \tag{10}$$

The independent unit-power streams give desired and interfering powers $|\mathbf{h}^H\mathbf{F}_a\mathbf{f}_c|^2$ and $|\mathbf{h}^H\mathbf{F}_a\mathbf{f}_s|^2$, respectively. Thus,

$$\mathrm{SINR}_c = \frac{|\mathbf{h}^H(\mathbf{p})\mathbf{F}_a\mathbf{f}_c|^2}{|\mathbf{h}^H(\mathbf{p})\mathbf{F}_a\mathbf{f}_s|^2 + \sigma_c^2}, \qquad \gamma_c = \log(1 + \mathrm{SINR}_c). \tag{11}$$

There is no inter-user interference. As in the reference model, the user treats the dedicated sensing stream as interference. We use natural logarithms throughout, giving the communication metric in nats/s/Hz; replacing both communication and sensing logarithms by $\log_2$ scales both metrics by the same constant. The standard achievable-rate interpretation assumes suitable Gaussian signaling and interference treatment.

### 1.6 Radar echoes and sensing metric

Index the target by $l = 0$ and clutter sources by $l = 1, \ldots, J$. Retain the bounded steering-response model of [1]:

$$\mathbf{a}_{t,l}(\mathbf{p}) = \widehat{\mathbf{a}}_t(\theta_l; \mathbf{p}) + \boldsymbol{\delta}_{t,l}, \tag{12}$$

$$\mathbf{a}_{r,l} = \widehat{\mathbf{a}}_r(\theta_l) + \boldsymbol{\delta}_{r,l}. \tag{13}$$

The round-trip channel becomes

$$\mathbf{H}_l(\mathbf{p}) = \alpha_l \mathbf{a}_{r,l} \mathbf{a}_{t,l}^T(\mathbf{p}) \in \mathbb{C}^{N_r \times N_t}, \qquad l = 0, \ldots, J, \tag{14}$$

where $\alpha_l$ contains the reflection and round-trip propagation gain. The receive ULA and the transpose convention are unchanged from the reference. The echo vector and filter output are

$$\mathbf{y}_r = \mathbf{H}_0(\mathbf{p})\mathbf{x} + \sum_{j=1}^{J} \mathbf{H}_j(\mathbf{p})\mathbf{x} + \mathbf{n}_r, \tag{15}$$

$$y_s = \mathbf{w}^H \mathbf{y}_r, \qquad \mathbf{n}_r \sim \mathcal{CN}(\mathbf{0}, \sigma_r^2 \mathbf{I}_{N_r}), \quad \|\mathbf{w}\|_2^2 = 1. \tag{16}$$

The effective echo model neglects residual transmitter self-interference and does not explicitly resolve delay or Doppler. The desired target power is

$$P_0 = \mathbb{E}|\mathbf{w}^H \mathbf{H}_0 \mathbf{x}|^2 = \mathbf{w}^H \mathbf{H}_0 \mathbf{R}_x \mathbf{H}_0^H \mathbf{w} \tag{17}$$

$$= \|\mathbf{w}^H \mathbf{H}_0 \mathbf{F}_a \mathbf{F}_d\|_2^2 = |\mathbf{w}^H \mathbf{H}_0 \mathbf{v}_c|^2 + |\mathbf{w}^H \mathbf{H}_0 \mathbf{v}_s|^2. \tag{18}$$

Both streams therefore illuminate the target. Retaining the reference's sum-of-powers treatment of effective clutter contributions gives

$$\mathrm{SINR}_s = \frac{\|\mathbf{w}^H \mathbf{H}_0(\mathbf{p}) \mathbf{F}_a \mathbf{F}_d\|_2^2}{\sum_{j=1}^{J} \|\mathbf{w}^H \mathbf{H}_j(\mathbf{p}) \mathbf{F}_a \mathbf{F}_d\|_2^2 + \sigma_r^2 \|\mathbf{w}\|_2^2}, \qquad \gamma_s = \log(1 + \mathrm{SINR}_s). \tag{19}$$

Here $\gamma_s$ is the inherited MI-based sensing metric, rather than a new claim about exact parameter-estimation mutual information. The expression retains the reference approximation that effective echo cross-correlations are negligible. Independent source streams alone do not guarantee this: coherent clutter would additionally contribute terms $\mathbf{w}^H \mathbf{H}_j \mathbf{R}_x \mathbf{H}_k^H \mathbf{w}$ for $j \neq k$. We preserve the original metric to isolate the effect of fluid positioning.

### 1.7 Uncertainty sets

Keep the uncertainty balls unchanged in form:

$$\mathcal{A}_h = \{\boldsymbol{\delta}_h \in \mathbb{C}^{N_t} : \|\boldsymbol{\delta}_h\|_2 \leq \epsilon_h \sqrt{N_t}\}, \tag{20}$$

$$\mathcal{A}_t = \prod_{l=0}^{J} \{\boldsymbol{\delta}_{t,l} \in \mathbb{C}^{N_t} : \|\boldsymbol{\delta}_{t,l}\|_2 \leq \epsilon_t \sqrt{N_t}\}, \tag{21}$$

$$\mathcal{A}_r = \prod_{l=0}^{J} \{\boldsymbol{\delta}_{r,l} \in \mathbb{C}^{N_r} : \|\boldsymbol{\delta}_{r,l}\|_2 \leq \epsilon_r \sqrt{N_r}\}. \tag{22}$$

Write $\boldsymbol{\Delta}_t = \{\boldsymbol{\delta}_{t,l}\}_{l=0}^{J}$ and $\boldsymbol{\Delta}_r = \{\boldsymbol{\delta}_{r,l}\}_{l=0}^{J}$. The radii are specified model parameters in the initial comparison. They are kept identical for fixed and fluid arrays. These are unstructured response-error sets; their elements need not correspond to a single physical angle error. No separate antenna-position error is introduced at this stage.

### 1.8 Joint robust optimization

For a prescribed tradeoff $0 < \rho < 1$, the proposed extension is

$$\begin{aligned} \underset{\mathbf{p},\mathbf{F}_a,\mathbf{F}_d,\mathbf{w}}{\text{maximize}} \quad & \rho \min_{\boldsymbol{\delta}_h \in \mathcal{A}_h} \gamma_c + (1-\rho) \min_{\substack{\boldsymbol{\Delta}_t \in \mathcal{A}_t \\ \boldsymbol{\Delta}_r \in \mathcal{A}_r}} \gamma_s \\ \text{subject to} \quad & |[\mathbf{F}_a]_{n,\ell}| = 1, \quad \forall n, \ell, \\ & \|\mathbf{F}_a \mathbf{F}_d\|_F^2 = P_{\text{BS}}, \qquad \|\mathbf{w}\|_2^2 = 1, \\ & \mathbf{p} \in \mathcal{P}. \end{aligned} \tag{23}$$

Relative to the single-user, single-target specialization of [1], the only new design block is $\mathbf{p}$. The received-signal formulas, uncertainty sets, hybrid constraints, and weighted robust objective retain their original structure. Position dependence enters through (4) and (14). Fixing $p_n = (n-1)d$ recovers the fixed-array geometry under the same spatial channel construction. This does not by itself reproduce an unrelated position-independent channel-generation distribution.

## A Additional Derivatives Introduced by Fluid Positions

The following derivatives hold for fixed beamformers and fixed additive uncertainty realizations. Let $\mathbf{e}_n$ be the $n$th coordinate vector and let a prime denote differentiation with respect to real coordinate $p_n$. From (2),

$$\mathbf{a}'_{t,l,n} = -jk_0 \sin\theta_l \, [\widehat{\mathbf{a}}_t(\theta_l; \mathbf{p})]_n \mathbf{e}_n. \tag{24}$$

Only the nominal response is differentiated, because the additive errors are held fixed in this partial derivative. Similarly,

$$(\mathbf{h}^H)'_n = \sum_{k=1}^{K_c} \beta_k (-jk_0 \sin\vartheta_k) e^{-jk_0 p_n \sin\vartheta_k} \mathbf{e}_n^T, \tag{25}$$

$$\mathbf{H}'_{l,n} = \alpha_l \mathbf{a}_{r,l} (\mathbf{a}'_{t,l,n})^T. \tag{26}$$

For any differentiable complex scalar $b(p_n)$,

$$\frac{\partial |b|^2}{\partial p_n} = 2\,\text{Re}\{b^* b'\}. \tag{27}$$

Define $b_c = \mathbf{h}^H \mathbf{v}_c$, $b_s = \mathbf{h}^H \mathbf{v}_s$, $N_c = |b_c|^2$, and $D_c = |b_s|^2 + \sigma_c^2$. Then

$$N'_{c,n} = 2\,\text{Re}\{b_c^* (\mathbf{h}^H)'_n \mathbf{v}_c\}, \tag{28}$$

$$D'_{c,n} = 2\,\text{Re}\{b_s^* (\mathbf{h}^H)'_n \mathbf{v}_s\}, \tag{29}$$

$$\frac{\partial \gamma_c}{\partial p_n} = \frac{D_c N'_{c,n} - N_c D'_{c,n}}{D_c (D_c + N_c)}. \tag{30}$$

For sensing, put $\mathbf{q}_l = \mathbf{w}^H \mathbf{H}_l \mathbf{F}_a \mathbf{F}_d$ and $\mathbf{q}'_{l,n} = \mathbf{w}^H \mathbf{H}'_{l,n} \mathbf{F}_a \mathbf{F}_d$. Define $N_s = \|\mathbf{q}_0\|_2^2$ and $D_s = \sum_{j=1}^J \|\mathbf{q}_j\|_2^2 + \sigma_r^2 \|\mathbf{w}\|_2^2$. It follows that

$$N'_{s,n} = 2\,\text{Re}\{\mathbf{q}'_{0,n} \mathbf{q}_0^H\}, \tag{31}$$

$$D'_{s,n} = 2\,\text{Re}\left\{ \sum_{j=1}^J \mathbf{q}'_{j,n} \mathbf{q}_j^H \right\}, \tag{32}$$

$$\frac{\partial \gamma_s}{\partial p_n} = \frac{D_s N'_{s,n} - N_s D'_{s,n}}{D_s (D_s + N_s)}. \tag{33}$$

Thus the fixed-error outer utility gradient is

$$\frac{\partial \gamma}{\partial p_n} = \rho \frac{\partial \gamma_c}{\partial p_n} + (1-\rho)\frac{\partial \gamma_s}{\partial p_n}. \tag{34}$$

For base-two logarithms, divide these utility derivatives by $\ln 2$. Using an approximately minimized error realization in these expressions gives the position-update direction of an alternating PGDA scheme; it is not an automatic certificate of the exact derivative of the globally minimized robust objective. The inherited beamforming/error gradients keep their algebraic structure, evaluated with the position-dependent channels.

# References


[1] L. Wang, S. A. Vorobyov, and E. Ollila, "L2O robust hybrid beamforming for ISAC," *IEEE Transactions on Wireless Communications*, 2026, doi: 10.1109/TWC.2026.3702925.

[2] L. Zhou, J. Yao, M. Jin, T. Wu, and K.-K. Wong, "Fluid antenna-assisted ISAC systems," *IEEE Wireless Communications Letters*, vol. 13, no. 12, 2024, doi: 10.1109/LWC.2024.3476148.